



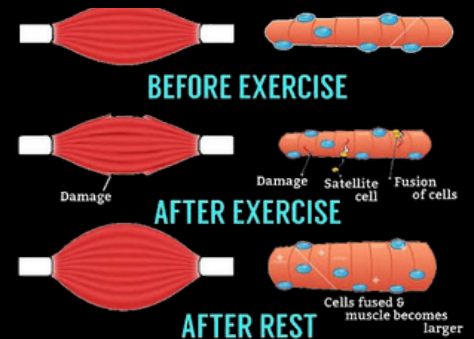
MUSCLE BUILDING

101

**ECHELON FITNESS
COLLECTIVE**



SCIENCE & SYSTEMS



WHAT IS MUSCLE & HOW IT WORKS

MUSCLE IS ACTIVE TISSUE THAT REQUIRES PHYSICAL STIMULUS TO GROW. YOUR BODY WILL NOT BUILD IT UNLESS IT IS FORCED TO ADAPT TO A DEMANDING ENVIRONMENT. GROWTH HAPPENS IN A SIMPLE THREE-STEP LOOP:

- **MECHANICAL TENSION (THE STIMULUS):** WHEN YOU LIFT HEAVY WEIGHTS, YOU CREATE MICROSCOPIC TEARS IN YOUR MUSCLE FIBERS. THIS ISN'T DAMAGE TO WORRY ABOUT—IT'S THE SIGNAL YOUR BODY NEEDS.
- **THE REPAIR WINDOW (HYPERTROPHY):** WHILE YOU REST AND SLEEP, YOUR BODY ACTIVATES AN IMMUNE RESPONSE TO REPAIR THOSE MICRO-TEARS, FUSING THE MUSCLE FIBERS BACK TOGETHER.
- **THE ADAPTATION:** THE BODY DOESN'T JUST FIX THE FIBER; IT BUILDS IT BACK THICKER AND STRONGER THAN BEFORE SO IT CAN HANDLE THAT SAME WEIGHT EASIER NEXT TIME.

*****THE GOLDEN RULE: IF YOU LIFT THE EXACT SAME WEIGHTS FOR THE EXACT SAME REPS EVERY WEEK, YOUR MUSCLE HAS NO REASON TO GROW. YOU MUST PRACTICE PROGRESSIVE OVERLOAD (ADDING WEIGHT, REPS, OR CONTROL OVER TIME).**

FUELING THE BUILD

WHAT TO EAT: THE HYPERTROPHY DIET



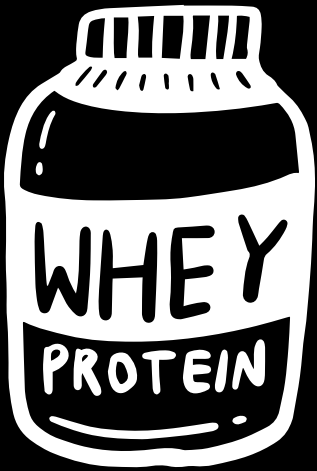
YOU CANNOT BUILD A HOUSE WITHOUT BRICKS AND WORKERS. IF YOUR GOAL IS BUILDING MUSCLE, YOUR NUTRITION MUST MEET TWO NON-NEGOTIABLE TARGETS:

- **THE BRICKS (PROTEIN):** PROTEIN BREAKS DOWN INTO AMINO ACIDS, WHICH ARE THE LITERAL BUILDING BLOCKS OF MUSCLE TISSUE.
- **THE TARGET:** EAT 0.8 TO 1.0 GRAMS OF PROTEIN PER POUND OF BODY WEIGHT DAILY (E.G., A 180-POUND PERSON NEEDS 144-180G OF PROTEIN).
- **SOURCES:** CHICKEN, LEAN BEEF, TURKEY, EGGS, FISH, TOFU, AND COTTAGE CHEESE.
- **THE ENERGY (THE CALORIC SURPLUS):** BUILDING NEW TISSUE REQUIRES EXTRA ENERGY. TO OPTIMIZE GROWTH WITHOUT GAINING EXCESS FAT, YOU NEED A "LEAN GAIN" SURPLUS.
- **THE TARGET:** EAT 200-300 CALORIES ABOVE YOUR MAINTENANCE LEVEL.
- **THE FUEL:** PRIORITIZE CLEAN CARBOHYDRATES (RICE, OATS, SWEET POTATOES) TO FUEL BRUTAL WORKOUTS, AND HEALTHY FATS (AVOCADO, NUTS) FOR HORMONE REGULATION.

SUPPLEMENTS THAT MATTER

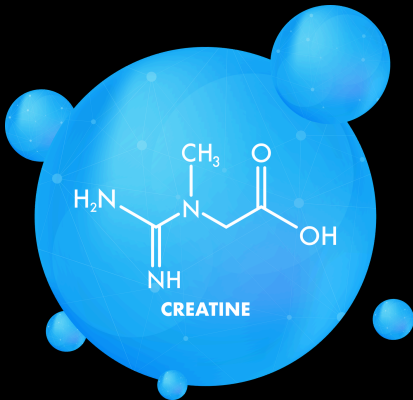
*ONLY TWO SUPPLEMENTS ARE UNIVERSALLY BACKED BY
DECADES OF DATA FOR BUILDING MUSCLE*

I. WHEY PROTEIN (THE CONVENIENCE KING)



- **WHAT** : A FAST-DIGESTING, HIGH-QUALITY PROTEIN DERIVED FROM MILK.
- **WHY** : IT IS INCREDIBLY DIFFICULT FOR MOST CLIENTS TO HIT THEIR DAILY PROTEIN TARGETS THROUGH WHOLE FOOD ALONE. WHEY PROVIDES A HIGHLY BIOAVAILABLE DOSE OF LEUCINE (THE AMINO ACID THAT FLIPS THE "GROWTH SWITCH" IN MUSCLES) WITHOUT THE EXTRA FAT OR PREP TIME.
- **HOW** : 1-2 SCOOPS DAILY, IDEALLY POST-WORKOUT OR AS A SNACK TO BRIDGE LONG GAPS BETWEEN MEALS.

II. CREATINE MONOHYDRATE (THE POWER PLANT)



- **WHAT** : A NATURALLY OCCURRING COMPOUND STORED IN YOUR MUSCLES THAT HELPS PRODUCE RAPID CELLULAR ENERGY (ATP).
- **WHY IT'S IMPORTANT**: CREATINE ACTS LIKE A BACKUP BATTERY FOR YOUR MUSCLES. IT ALLOWS YOU TO SQUEEZE OUT 1-2 MORE HEAVY REPS ON YOUR LIFTS, WHICH ADDS UP TO MASSIVE VOLUME OVER TIME. IT ALSO PULLS WATER INSIDE THE MUSCLE CELLS (INTRACELLULAR HYDRATION), MAKING THEM APPEAR FULLER AND INCREASING NUTRIENT SIGNALING.
- **HOW** : TAKE 5 GRAMS EVERY SINGLE DAY AT ANY TIME. CONSISTENCY IS KEY; IT TAKES ABOUT 2-3 WEEKS TO FULLY SATURATE YOUR MUSCLES. (NO NEED TO CYCLE OFF IT).